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Class : MCA - I.

Subject : python programming

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Type : CCE - II.

- 1) write a program that ,
 a) write a python program using a lambda function.
 to find the square of a number.

code:

```
square = lambda x : x * x
num = int(input("Enter a number:"))
result = square(num)
print("Square of", num, "is :", result)
```

Output:

Enter a number : 5

Square of 5 is : 25

- b) Create simple decorator that prints function.
 started. Before executing a function.

code:

```
def start_decorator(func):
```

```
    def wrapper():
```

```
        print("Function started")
```

```
        func()
```

```
    return wrapper
```

@start_decorator.

```
def greet():
```

```
    print("Hello world")
```

```
greet()
```

output:

function started.

hello ! world.

c) write a generator function that yields first 5 even numbers:

code:

```
def first_five_even():  
    for i in range(1, 6):  
        yield i*2
```

```
for num in first_five_even():  
    print(num).
```

output:

2

4

6

8

10

2) write a python program that

a) Define a function outer.

b) inside it, defines a nested function inner()

c) use the nonlocal keyword to modify a variable from the outer function.

d) displays the updated value.

code:

```
def outer():  
    x = 10  
    print("Before modification:", x)  
    def inner():  
        nonlocal x
```

```
x = 20
inner()
print("After modification :", x)
```

```
outer()
```

output:

Before modification: 10.

After modification: 20.

3) write a program that,

- Uses a lambda function to find the cube of a number
- Define a decorator that prints Executing Function before calling a function.
- Define a generator to generate first 5 odd numbers

code:

```
cube = lambda x: x**3
num = int(input("Enter a number: "))
print("Cube of", num, "is:", cube(num))
```

```
def my_decorator(func):
```

```
    def wrapper(*args, **kwargs):
```

```
        print("Executing function")
```

```
        return func(*args, **kwargs)
```

```
    return wrapper
```

```
@my_decorator
```

```
def greet():
```

```
    print("Hello from decorated function!")
```

```
greet()
```



```
def first_five_odd():
    count = 0
    num = 1
    while count < 5:
        yield num
        num += 1
        count += 1
    print("first 5 odd numbers:").
```

```
for i in first_five_odd():
    print(i).
```

output:

Enter a number: 5

Cube of 5 is: 125

Executing function.

Hello from decorated function!

first 5 odd numbers:

1

3

5

7

9

4) write a python program that,

a) Imports the math module.

b) Calculates factorial of a number using math module.

code:

```
import math.
```

```
num = int(input("Enter a number")).
```

```
fact = math.factorial(num)
```

```
print("Factorial of", num, "is:", fact)
```

output:

Enter a number: 5

factorial of 5 is: 120.

5) write a python program that (Use, random, datetime and math modules).

a) Generates a random number between 10 and 50.

b) Displays the current date and time.

c) calculates square root of 144.

code:

```
import random.
```

```
import datetime.
```

```
import math.
```

```
random_number = random.randint(10, 50).
```

```
print("Random number between 10 and 50",  
      random_number).
```

```
current_datetime = datetime.datetime.now().
```

```
print("Current Date and Time", current_datetime)
```

```
sqrt_value = math.sqrt(144).
```

```
print("Square root of 144 is:", sqrt_value).
```

output:

random number between 10 and 50: 40.

Current date and time: 2026-03-02T21:11:10.4593

Square root of 144 is: 12.0.

6) write a python program that,

a) import a user-defined module (assume module: mymodule)

b) Uses regular expression to validate a password and mobile number with your rules.

code :

```
import mymodule
import re
```

```
my module display message()
password = input("Enter password")
password-pattern = r"^(?=.*[a-z])(?=.*[A-Z])
(.*\d)(?=.*[@#$_])
[A-Za-z\d@#$_]{8,}$"
if re.match(password-pattern, password):
    print("valid password")
else:
    print("Invalid password")
mobile = input("Enter mobile number")
mobile-pattern = r"^[6-9]\d{9}$"
if re.match(mobile-pattern, mobile):
    print("valid mobile number")
else:
    print("Invalid mobile number")
```

output :

```
User-defined module imported successfully
Enter password : 7ed@12345
valid password
Enter mobile number: 9894189398
valid mobile number
```